RAG-Based Business Report Analysis — Full-Code Submission

Business Context

As organizations grow and scale, they are often inundated with large volumes of data, reports, and documents that contain critical information for decision-making. In real-world business settings, such as venture capital firms like Andreesen Horowitz, business analysts are required to sift through large datasets, research papers, or reports to extract relevant information that impacts strategic decisions.

For instance, consider that you've just joined Andreesen Horowitz, a renowned venture capital firm, and you are tasked with analyzing a dense report like the Harvard Business Review's **"How Apple is Organized for Innovation."** Going through the report manually can be extremely time-consuming as the size and complexity of these report increases. However, by using **Semantic Search** and **Retrieval-Augmented Generation (RAG)** models, you can significantly streamline this process.

Imagine having the capability to directly ask questions like, “How does Apple structure its teams for innovation?” and get immediate, relevant answers drawn from the report. This ability to extract and organize specific insights quickly and accurately enables you to focus on higher-level analysis and decision-making, rather than being bogged down by information retrieval.

OpenAI API Setup

This notebook uses the OpenAI Responses API for answer generation. Store the key in Google Colab Secrets as `OPENAI_API_KEY` ; do not paste the key directly into a code cell.

The course-recommended T4 GPU accelerates local embeddings, while OpenAI generation runs through the hosted API.

Objective

The goal is to develop a RAG application that helps business analysts efficiently extract key insights from extensive reports, such as “How Apple is Organized for Innovation.”

Specifically, the system aims to:

- Answer user queries by retrieving relevant content directly from lengthy documents.
- Support natural-language interaction without requiring a full manual read-through.
- Act as an intelligent assistant that streamlines the report analysis process.

Through this solution, analysts can save time, improve productivity, and make faster, more informed strategic decisions

Data Description

How Apple is Organized for Innovation - An article of 11 pages in pdf format

Installing and Importing Necessary Libraries and Dependencies

In [ ]:
```
# Install the libraries required for the OpenAI-based RAG pipeline.
!pip install \
    openai \
    langchain_community==0.3.27 \
    langchain==0.3.27 \
    chromadb==1.0.15 \
    pymupdf==1.26.3 \
    tiktoken==0.9.0 \
    datasets==4.0.0 \
    evaluate==0.4.5 \
    sentence-transformers==3.4.1
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Requirement already satisfied: openai in /usr/local/lib/python3.12/dis
t-packages (2.45.0)
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Collecting sentence-transformers==3.4.1
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Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/py
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al/lib/python3.12/dist-packages (from langchain_community==0.3.27) (9.
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14_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl.metadata (8.7
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  Downloading posthog-5.4.0-py3-none-any.whl.metadata (5.7 kB)
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ib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_com
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Collecting marshmallow<4.0.0,>=3.18.0 (from dataclasses-json<0.7,>=0.
5.7->langchain_community==0.3.27)
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chain_community==0.3.27) (1.33)
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ib/python3.12/dist-packages (from langchain-core<1.0.0,>=0.3.66->langc
hain_community==0.3.27) (0.17.0)
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nity==0.3.27) (1.0.0)
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Collecting opentelemetry-exporter-otlp-proto-common==1.44.0 (from open
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15)
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b/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain==0.
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n3.12/dist-packages (from torch>=1.11.0->sentence-transformers==3.4.1)
(3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dis
t-packages (from torch>=1.11.0->sentence-transformers==3.4.1) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/
python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1; platfo
rm_system == "Linux"->torch>=1.11.0->sentence-transformers==3.4.1) (1
2.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/loca
l/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transform
ers==3.4.1) (12.9.7)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/lo
cal/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transfo
rmers==3.4.1) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/l
ocal/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transf
ormers==3.4.1) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/
lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformer
s==3.4.1) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/loca
l/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transform
ers==3.4.1) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python
3.12/dist-packages (from torch>=1.11.0->sentence-transformers==3.4.1)
(3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/
local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cu
fft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.
1; platform_system == "Linux"->torch>=1.11.0->sentence-transformers==
3.4.1) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in
/usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda
rt,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==
12.8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transforme
rs==3.4.1) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/
local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cu
fft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.
1; platform_system == "Linux"->torch>=1.11.0->sentence-transformers==
3.4.1) (11.3.3.83)
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Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/
local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cu
fft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.
1; platform_system == "Linux"->torch>=1.11.0->sentence-transformers==
3.4.1) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /u
sr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudar
t,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==1
2.8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformer
s==3.4.1) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /us
r/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,
cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.
8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformers=
=3.4.1) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /u
sr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudar
t,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==1
2.8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformer
s==3.4.1) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /u
sr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudar
t,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==1
2.8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformer
s==3.4.1) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /us
r/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,
cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.
8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformers=
=3.4.1) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /u
sr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudar
t,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==1
2.8.1; platform_system == "Linux"->torch>=1.11.0->sentence-transformer
s==3.4.1) (12.8.93)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/loc
al/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cuff
t,cufile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==12.8.1;
platform_system == "Linux"->torch>=1.11.0->sentence-transformers==3.4.
1) (12.8.90)
Collecting huggingface-hub>=0.24.0 (from datasets==4.0.0)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/py
thon3.12/dist-packages (from transformers<5.0.0,>=4.41.0->sentence-tra
nsformers==3.4.1) (0.8.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/py
thon3.12/dist-packages (from typer>=0.9.0->chromadb==1.0.15) (1.5.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/
python3.12/dist-packages (from typer>=0.9.0->chromadb==1.0.15) (0.0.4)
Requirement already satisfied: httptools>=0.8.0 in /usr/local/lib/pyth
on3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb==1.0.1
5) (0.8.0)
Requirement already satisfied: uvloop>=0.15.1 in /usr/local/lib/python
3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb==1.0.15)
(0.22.1)
Requirement already satisfied: watchfiles>=0.20 in /usr/local/lib/pyth
on3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb==1.0.1
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5) (1.2.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.
12/dist-packages (from pandas->datasets==4.0.0) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python
3.12/dist-packages (from pandas->datasets==4.0.0) (2026.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python
3.12/dist-packages (from scikit-learn->sentence-transformers==3.4.1)
(1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/
python3.12/dist-packages (from scikit-learn->sentence-transformers==3.
4.1) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/
python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=1.11.
0->sentence-transformers==3.4.1) (1.5.6)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/pyth
on3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langchain-core<1.
0.0,>=0.3.66->langchain_community==0.3.27) (3.1.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.1
2/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->chromadb==
1.0.15) (0.1.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/py
thon3.12/dist-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-tr
ansformers==3.4.1) (1.3.0)
Collecting mypy-extensions>=0.3.0 (from typing-inspect<1,>=0.4.0->data
classes-json<0.7,>=0.5.7->langchain_community==0.3.27)
  Downloading mypy_extensions-1.1.0-py3-none-any.whl.metadata (1.1 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/pytho
n3.12/dist-packages (from jinja2->torch>=1.11.0->sentence-transformers
==3.4.1) (3.0.3)
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/pytho
n3.12/dist-packages (from requests-oauthlib->kubernetes>=28.1.0->chrom
adb==1.0.15) (3.3.1)
Downloading langchain_community-0.3.27-py3-none-any.whl (2.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.5/2.5 MB 48.1 MB/s eta
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Downloading langchain-0.3.27-py3-none-any.whl (1.0 MB)
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Downloading chromadb-1.0.15-cp39-abi3-manylinux_2_17_x86_64.manylinux2
014_x86_64.whl (19.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 19.5/19.5 MB 57.4 MB/s eta
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Downloading pymupdf-1.26.3-cp39-abi3-manylinux_2_28_x86_64.whl (24.1 M
B)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 24.1/24.1 MB 68.8 MB/s eta
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Downloading tiktoken-0.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux
2014_x86_64.whl (1.2 MB)
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Downloading evaluate-0.4.5-py3-none-any.whl (84 kB)
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Downloading sentence_transformers-3.4.1-py3-none-any.whl (275 kB)
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ta 0:00:00
Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
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Downloading build-1.5.0-py3-none-any.whl (26 kB)
Downloading dataclasses_json-0.6.7-py3-none-any.whl (28 kB)
Downloading httpx_sse-0.4.3-py3-none-any.whl (9.0 kB)
Downloading kubernetes-36.0.3-py2.py3-none-any.whl (4.6 MB)
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Downloading langchain_core-0.3.86-py3-none-any.whl (461 kB)
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ta 0:00:00
Downloading langchain_text_splitters-0.3.11-py3-none-any.whl (33 kB)
Downloading onnxruntime-1.27.0-cp312-cp312-manylinux_2_27_x86_64.manyl
inux_2_28_x86_64.whl (18.7 MB)
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Downloading opentelemetry_exporter_otlp_proto_grpc-1.44.0-py3-none-an
y.whl (19 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.44.0-py3-none-a
ny.whl (17 kB)
Downloading opentelemetry_proto-1.44.0-py3-none-any.whl (72 kB)
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Downloading opentelemetry_sdk-1.44.0-py3-none-any.whl (137 kB)
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ta 0:00:00
Downloading opentelemetry_api-1.44.0-py3-none-any.whl (60 kB)
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Downloading opentelemetry_semantic_conventions-0.65b0-py3-none-any.whl
(204 kB)
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Downloading overrides-7.7.0-py3-none-any.whl (17 kB)
Downloading packaging-25.0-py3-none-any.whl (66 kB)
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Downloading posthog-5.4.0-py3-none-any.whl (105 kB)
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ta 0:00:00
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014
_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (71 kB)
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Downloading pydantic_settings-2.14.2-py3-none-any.whl (61 kB)
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Downloading pypika-0.51.1-py2.py3-none-any.whl (60 kB)
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Downloading transformers-4.57.6-py3-none-any.whl (12.0 MB)
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Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
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ta 0:00:00
Downloading backoff-2.2.1-py3-none-any.whl (15 kB)
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Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading marshmallow-3.26.2-py3-none-any.whl (50 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.0/51.0 kB 4.6 MB/s eta
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Downloading typing_inspect-0.9.0-py3-none-any.whl (8.8 kB)
Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Downloading mypy_extensions-1.1.0-py3-none-any.whl (5.0 kB)
Installing collected packages: pypika, durationpy, pyproject_hooks, py
mupdf, pybase64, packaging, overrides, opentelemetry-proto, openteleme
try-api, mypy-extensions, httpx-sse, bcrypt, backoff, typing-inspect,
tiktoken, posthog, opentelemetry-semantic-conventions, opentelemetry-e
xporter-otlp-proto-common, onnxruntime, marshmallow, huggingface-hub,
build, pydantic-settings, opentelemetry-sdk, kubernetes, dataclasses-j
son, transformers, opentelemetry-exporter-otlp-proto-grpc, langchain-c
ore, sentence-transformers, langchain-text-splitters, evaluate, chroma
db, langchain, langchain_community
  Attempting uninstall: packaging
    Found existing installation: packaging 26.2
    Uninstalling packaging-26.2:
      Successfully uninstalled packaging-26.2
  Attempting uninstall: opentelemetry-api
    Found existing installation: opentelemetry-api 1.42.1
    Uninstalling opentelemetry-api-1.42.1:
      Successfully uninstalled opentelemetry-api-1.42.1
  Attempting uninstall: tiktoken
    Found existing installation: tiktoken 0.13.0
    Uninstalling tiktoken-0.13.0:
      Successfully uninstalled tiktoken-0.13.0
  Attempting uninstall: opentelemetry-semantic-conventions
    Found existing installation: opentelemetry-semantic-conventions 0.
63b1
    Uninstalling opentelemetry-semantic-conventions-0.63b1:
      Successfully uninstalled opentelemetry-semantic-conventions-0.63
b1
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.23.0
    Uninstalling huggingface_hub-1.23.0:
      Successfully uninstalled huggingface_hub-1.23.0
  Attempting uninstall: opentelemetry-sdk
    Found existing installation: opentelemetry-sdk 1.42.1
    Uninstalling opentelemetry-sdk-1.42.1:
      Successfully uninstalled opentelemetry-sdk-1.42.1
  Attempting uninstall: transformers
    Found existing installation: transformers 5.13.1
    Uninstalling transformers-5.13.1:
      Successfully uninstalled transformers-5.13.1
  Attempting uninstall: langchain-core
    Found existing installation: langchain-core 1.4.9
    Uninstalling langchain-core-1.4.9:
      Successfully uninstalled langchain-core-1.4.9
  Attempting uninstall: sentence-transformers
    Found existing installation: sentence-transformers 5.6.0
    Uninstalling sentence-transformers-5.6.0:
      Successfully uninstalled sentence-transformers-5.6.0
  Attempting uninstall: langchain
    Found existing installation: langchain 1.3.13
    Uninstalling langchain-1.3.13:
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      Successfully uninstalled langchain-1.3.13
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of t
he following dependency conflicts.
langgraph-prebuilt 1.1.0 requires langchain-core>=1.3.1, but you have
langchain-core 0.3.86 which is incompatible.
langgraph-sdk 0.4.2 requires langchain-core<2,>=1.4.0, but you have la
ngchain-core 0.3.86 which is incompatible.
langgraph 1.2.9 requires langchain-core<2,>=1.4.7, but you have langch
ain-core 0.3.86 which is incompatible.
google-adk 2.4.0 requires opentelemetry-api<=1.42.1,>=1.39, but you ha
ve opentelemetry-api 1.44.0 which is incompatible.
google-adk 2.4.0 requires opentelemetry-sdk<=1.42.1,>=1.39, but you ha
ve opentelemetry-sdk 1.44.0 which is incompatible.
gradio 6.20.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggi
ngface-hub 0.36.2 which is incompatible.
Successfully installed backoff-2.2.1 bcrypt-5.0.0 build-1.5.0 chromadb
-1.0.15 dataclasses-json-0.6.7 durationpy-0.10 evaluate-0.4.5 httpx-ss
e-0.4.3 huggingface-hub-0.36.2 kubernetes-36.0.3 langchain-0.3.27 lang
chain-core-0.3.86 langchain-text-splitters-0.3.11 langchain_community-
0.3.27 marshmallow-3.26.2 mypy-extensions-1.1.0 onnxruntime-1.27.0 ope
ntelemetry-api-1.44.0 opentelemetry-exporter-otlp-proto-common-1.44.0
opentelemetry-exporter-otlp-proto-grpc-1.44.0 opentelemetry-proto-1.4
4.0 opentelemetry-sdk-1.44.0 opentelemetry-semantic-conventions-0.65b0
overrides-7.7.0 packaging-25.0 posthog-5.4.0 pybase64-1.4.3 pydantic-s
ettings-2.14.2 pymupdf-1.26.3 pypika-0.51.1 pyproject_hooks-1.2.0 sent
ence-transformers-3.4.1 tiktoken-0.9.0 transformers-4.57.6 typing-insp
ect-0.9.0
```

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in ***this notebook***.

In [ ]:
```
import os
import re
import json
import time
import textwrap
import warnings
from pathlib import Path

import pandas as pd
import matplotlib.pyplot as plt
import fitz
import torch

from openai import OpenAI
from langchain_core.documents import Document
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.embeddings import HuggingFaceEmbeddings
from langchain_community.vectorstores import Chroma

warnings.filterwarnings("ignore")

print("PyTorch version:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
    print(
        "GPU memory (GB):",
        round(torch.cuda.get_device_properties(0).total_memory / 1024**3, 2)
    )
else:
    print(
        "Warning: No GPU detected. OpenAI generation will still work, "
        "but local embeddings may run more slowly."
    )
```

```
PyTorch version: 2.11.0+cu128
CUDA available: True
GPU: Tesla T4
GPU memory (GB): 14.56
```

In [ ]:
```
print("pandas:", pd.__version__)
print("Path:", Path)
```

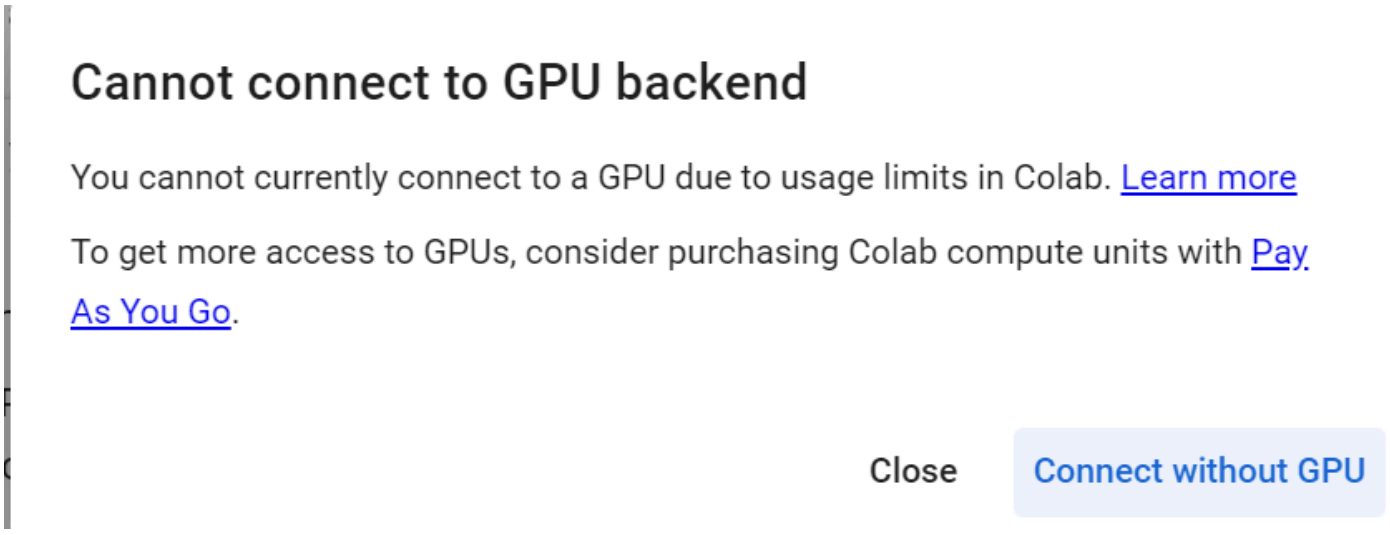
```
pandas: 2.2.2
Path: <class 'pathlib.Path'>
```

Question Answering using LLM

Note 1: When choosing between an open-source Hugging Face (HF) model and OpenAI's proprietary model, base your decision on your specific needs. If you opt for a Hugging Face model, make sure to connect to a GPU to execute the code efficiently.

Note 2: If the free-tier GPU of Google Colab is not accessible (due to unavailability or exhaustion of daily limit or other reasons), the following steps can be taken:

1. Wait for 12-24 hours until the GPU is accessible again or the daily usage limits are reset.
2. Switch to a different Google account and resume working on the project from there.
3. Try using the CPU runtime:
    - To use the CPU runtime, click on *Runtime => Change runtime type => CPU => Save*
    - One can also click on the *Continue without GPU* option to switch to a CPU runtime (kindly refer to the snapshot below)
    - The instructions for running the code on the CPU are provided in the respective sections of the notebook.



Downloading and Loading the model

In [ ]:

```python
# Add OPENAI_API_KEY to Google Colab Secrets instead of pasting it int
o code.
from openai import OpenAI
import json
from google.colab import userdata

client = OpenAI(
    base_url="https://aibe.mygreatlearning.com/openai/v1",
    api_key=userdata.get("OPENAI_API_KEY")
)

OPENAI_MODEL = "gpt-4o-mini"

MODEL_PARAMETERS = {
    "model": OPENAI_MODEL,
    "temperature": 0.0,
    "max_tokens": 700
}

print(json.dumps(MODEL_PARAMETERS, indent=2))

def generate_text(
    prompt: str,
    max_tokens: int = 700,
    temperature: float = 0.0
) -> str:
    response = client.chat.completions.create(
        model=OPENAI_MODEL,
        messages=[
            {
                "role": "user",
                "content": prompt
            }
        ],
        max_tokens=max_tokens,
        temperature=temperature
    )

    return response.choices[0].message.content.strip()
```

```
{
  "model": "gpt-4o-mini",
  "temperature": 0.0,
  "max_tokens": 700
}
```

Question 1: Who are the authors of this article and who published this article?

In [ ]:
```
question_1 = "Who are the authors of the article 'How Apple Is Organiz
ed for Innovation,' and who published it?"

baseline_prompt_1 = f"""[INST]
Answer the following question using only what you already know.
If you are uncertain, clearly say so.

Question: {question_1}
[/INST]"""

baseline_answer_1 = generate_text(baseline_prompt_1)
print(baseline_answer_1)
```

```
The article "How Apple Is Organized for Innovation" was authored by Jo
nny G. and published by Harvard Business Review.
```

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

In [ ]:
```
question_2 = """List the three leadership characteristics discussed in
'How Apple Is Organized for Innovation.' Use bullet points and explain
each characteristic in no more than two lines."""

baseline_prompt_2 = f"""[INST]
Answer the following question using only what you already know.
If you cannot verify the requested details, clearly state that limitat
ion.

Question: {question_2}
[/INST]"""

baseline_answer_2 = generate_text(baseline_prompt_2, max_tokens=550)
print(baseline_answer_2)
```

```
I cannot provide specific details from the document "How Apple Is Orga
nized for Innovation" as I do not have access to that content. Howeve
r, I can discuss general leadership characteristics that are often ass
ociated with innovative organizations. If you would like that, please
let me know!
```

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

In [ ]:
```
question_3 = """Explain specific examples from the article where Appl
e's
approach to leadership led to successful innovations."""

baseline_prompt_3 = f"""[INST]
Answer the following question using only what you already know.
Distinguish verified facts from assumptions and do not invent quotatio
ns.

Question: {question_3}
[/INST]"""

baseline_answer_3 = generate_text(baseline_prompt_3, max_tokens=650)
print(baseline_answer_3)
```

```
I don't have access to specific articles, but I can provide a general
overview of Apple's approach to leadership and how it has historically
led to successful innovations.

1. **Visionary Leadership**: Under Steve Jobs, Apple emphasized a clea
r vision for product design and user experience. This focus on aesthet
ics and functionality led to the creation of iconic products like the
iPhone and iPad, which revolutionized their respective markets.

2. **Cross-Functional Collaboration**: Apple's leadership encouraged c
ollaboration across different departments, such as design, engineerin
g, and marketing. This approach facilitated the development of product
s that were not only technologically advanced but also user-friendly,
as seen in the seamless integration of hardware and software in Apple
devices.

3. **Emphasis on Quality**: Apple's leadership has consistently priori
tized quality over quantity. This commitment is evident in the meticul
ous attention to detail in product design and manufacturing processes,
resulting in high-quality products that have garnered customer loyalt
y.

4. **Innovation Culture**: Apple fosters a culture of innovation where
employees are encouraged to think creatively and challenge the status
quo. This culture has led to groundbreaking innovations, such as the A
pp Store, which transformed how software is distributed and consumed.

These examples illustrate how Apple's leadership approach has been ins
trumental in driving successful innovations throughout its history.
```

Baseline LLM Observations

- The direct LLM responses are fast and readable, but the model has not been given the assigned article.
- Article-specific names, leadership principles, and innovation examples may therefore be incomplete or hallucinated.
- This establishes the need for source retrieval before generation.

Question Answering using LLM with Prompt Engineering

Question 1: Who are the authors of this article and who published this article?

In [ ]:
```
system_prompt = """You are a careful business-research assistant.
Answer only when you have reliable information. Do not fabricate name
s,
examples, quotations, or publication details. When the source article
is
not supplied, explicitly identify that limitation."""

prompt_engineered_1 = f"""
System instruction:
{system_prompt}

Task:
{question_1}

Return:
1. Authors
2. Publisher
3. A brief confidence statement
"""

pe_answer_1 = generate_text(prompt_engineered_1)
print(pe_answer_1)
```

```
I do not have access to specific articles or their details, including
the authors and publishers of "How Apple Is Organized for Innovation."
To find this information, I recommend checking a reliable academic dat
abase or the publication's official website.
```

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

In [ ]:
```
prompt_engineered_2 = f"""[INST]
System instruction:
{system_prompt}

Task:
{question_2}

Formatting requirements:
- Use exactly three bullet points.
- Name each characteristic.
- Keep each explanation to two lines or fewer.
- Do not substitute general leadership principles for article-specific
content.
[/INST]"""

pe_answer_2 = generate_text(prompt_engineered_2, max_tokens=550)
print(pe_answer_2)
```

```
I currently do not have access to the specific article "How Apple Is O
rganized for Innovation," so I cannot provide the three leadership cha
racteristics discussed in it. If you can provide the content or key po
ints from the article, I would be happy to help summarize or analyze t
hat information.
```

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

In [ ]:
```
prompt_engineered_3 = f"""[INST]
System instruction:
{system_prompt}

Task:
{question_3}

Formatting requirements:
- Identify concrete innovations or product features.
- Explain how leadership behavior contributed to each result.
- Do not claim that an example appears in the article unless certain.
- State any source limitation.
[/INST]"""

pe_answer_3 = generate_text(prompt_engineered_3, max_tokens=650)
print(pe_answer_3)
```

```
I currently do not have access to the specific article you are referri
ng to, so I cannot provide concrete examples of Apple's approach to le
adership and its impact on successful innovations. If you can provide
details or excerpts from the article, I would be happy to help analyze
those examples and explain how leadership behavior contributed to the
innovations mentioned. Please share any relevant information or contex
t.
```

Prompt-Engineering Observations

- Clear instructions improve structure, conciseness, and uncertainty disclosure.
- Prompt engineering alone cannot supply facts that are absent from the model prompt.
- The next stage adds the report itself through retrieval-augmented generation.

Data Preparation for RAG

The RAG pipeline uses the uploaded source file `HBR_How_Apple_Is_Organized_For_Innovation.pdf` .

Loading the Data

Required Source File

Upload the assigned article to the Google Colab working folder with the exact filename:

```
HBR_How_Apple_Is_Organized_For_Innovation.pdf
```

The notebook will stop with a clear error if that exact file is not present. All retrieval and RAG answers are generated from this PDF only.

In [ ]:

```
# Load the specific Harvard Business Review PDF from the Colab working
folder.
# Before running this cell, upload the file to Colab with the exact na
me:
# HBR_How_Apple_Is_Organized_For_Innovation.pdf

PDF_FILENAME = "HBR_How_Apple_Is_Organized_For_Innovation.pdf"
SOURCE_DOCUMENT = PDF_FILENAME
PDF_PATH = Path(PDF_FILENAME)

if not PDF_PATH.exists():
    raise FileNotFoundError(
        f"Required file not found: {PDF_FILENAME}\n\n"
        "Upload the PDF to the Google Colab working folder using the e
xact "
        "filename shown above, then rerun this cell."
    )

print("Using source document:", PDF_PATH.resolve())

# Extract text one page at a time so the page number is preserved
# for retrieval, citations, and answer verification.
pdf_document = fitz.open(PDF_PATH)
documents = []

for page_number, page in enumerate(pdf_document, start=1):
    page_text = page.get_text("text")
    page_text = re.sub(r"\s+", " ", page_text).strip()

    documents.append(
        Document(
            page_content=page_text,
            metadata={
                "source": PDF_FILENAME,
                "page": page_number
            }
        )
    )

pdf_document.close()

print(f"Loaded {len(documents)} pages from {PDF_FILENAME}.")
```

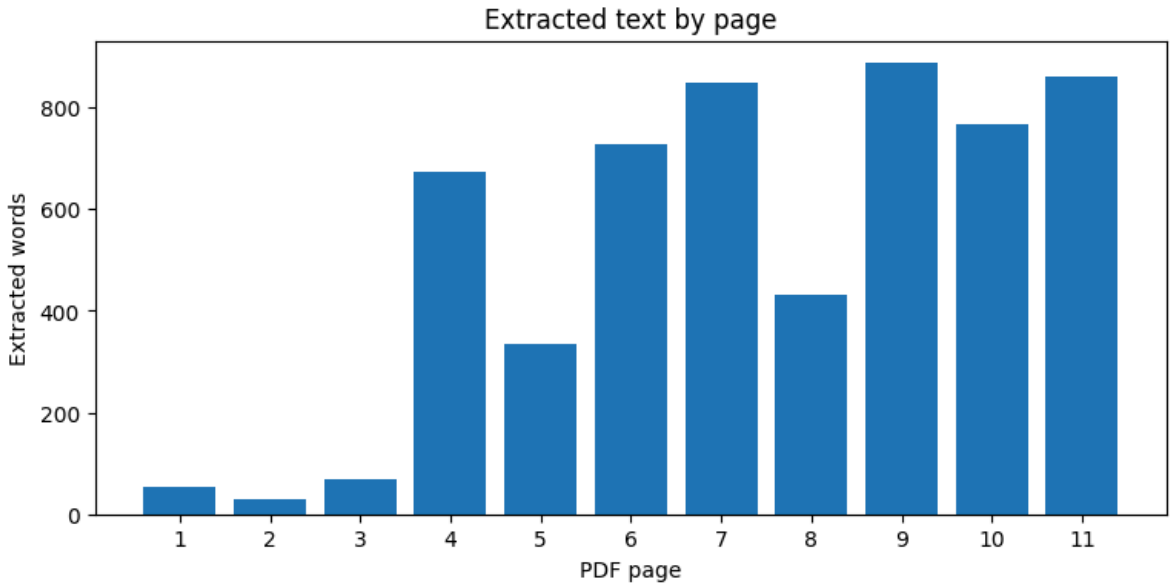
```
Using source document: /content/HBR_How_Apple_Is_Organized_For_Innovat
ion.pdf
Loaded 11 pages from HBR_How_Apple_Is_Organized_For_Innovation.pdf.
```

Data Overview

In [ ]:
```
# Create a concise overview of the source document.
overview_rows = []

for doc in documents:
    overview_rows.append({
        "page": doc.metadata["page"],
        "characters": len(doc.page_content),
        "words": len(doc.page_content.split()),
        "preview": doc.page_content[:140] + "..."
    })

overview_df = pd.DataFrame(overview_rows)
display(overview_df)

print("\nDocument totals")
print("Pages:", len(documents))
print("Characters:", overview_df["characters"].sum())
print("Words:", overview_df["words"].sum())

# Plot words extracted per page to identify sparse or image-heavy page
s.
plt.figure(figsize=(9, 4))
plt.bar(overview_df["page"], overview_df["words"])
plt.xlabel("PDF page")
plt.ylabel("Extracted words")
plt.title("Extracted text by page")
plt.xticks(overview_df["page"])
plt.show()

# Display a sample from the first content-heavy page.
sample_doc = max(documents, key=lambda doc: len(doc.page_content))
print(f"Sample from page {sample_doc.metadata['page']}:\n")
print(textwrap.fill(sample_doc.page_content[:1500], width=100))
```

	page	characters	words	preview
0	1	349	53	REPRINT R2006F PUBLISHED IN HBR NOVEMBER–DECEM...
1	2	198	30	2 Harvard Business Review November–December 20...
2	3	486	69	PHOTOGRAPHER MIKAEL JANSSON How Apple Is Organ...
3	4	4371	673	WELL KNOWN FOR ITS innovations in hardware, so...
4	5	2183	335	WHY A FUNCTIONAL ORGANIZATION? Apple's main pu...
5	6	4664	728	targets were the overriding criteria for judgi...
6	7	5305	849	specialization. Apple's leaders believe that w...
7	8	2717	431	40 specialist teams: silicon design, camera so...
8	9	5503	887	computational-photography techniques, Apple c...
9	10	4772	766	As the company has grown, entering new markets...
10	11	5533	861	collaborating across the many units. For insta...

```
Document totals
Pages: 11
Characters: 36081
Words: 5682
```



Sample from page 11:

collaborating across the many units. For instance, whereas Rosner is r
esponsible for the engineering
side of News, other managers oversee the operating system on which it
depends, the content, and the
business relationships with content creators (such as the New York Tim
es) and advertisers. To cope,
Rosner has adapted his role. As an expert who leads other experts, he
had been immersed in details—
especially those concerning the top-level aspects of software applicat
ions and their architecture
that affect how users engage with the software. He also collaborated w
ith manag- ers across the
company in projects that involved those areas. But with the expansion
of his responsibilities, he
has moved some things from his owning box—including tradi- tional prod
uctivity apps such as Keynote
and Pages—into his teaching box. (See the exhibit “Roger Rosner’s Disc
retionary Leadership.”) Now he
guides and gives feedback to other team members so that they can devel
op software applica- tions
according to Apple’s norms. Being a teacher doesn’t mean that Rosner g
ives instruction at a
whiteboard; rather, he offers strong, often passionate critiques of hi
s team’s work. (Clearly,
general managers without his core expertise would find it difficult to
teach what they don’t know.)
The second challenge for Rosner involved the addition of activities be
yond his original expertise.
Six years ago he was given responsibility for the engineering and desi
gn of News. Consequently, he
had to learn about publ

Data Chunking

In [ ]:
```
# Split pages into overlapping chunks.
# Character-based splitting is transparent and works well for this sho
rt report.
CHUNK_SIZE = 900
CHUNK_OVERLAP = 150

text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=CHUNK_SIZE,
    chunk_overlap=CHUNK_OVERLAP,
    separators=["\n\n", "\n", ". ", " ", ""]
)

chunks = text_splitter.split_documents(documents)

# Add stable chunk identifiers for traceability.
for chunk_id, chunk in enumerate(chunks):
    chunk.metadata["chunk_id"] = chunk_id

chunk_stats = pd.DataFrame({
    "chunk_id": [c.metadata["chunk_id"] for c in chunks],
    "page": [c.metadata["page"] for c in chunks],
    "characters": [len(c.page_content) for c in chunks],
    "words": [len(c.page_content.split()) for c in chunks]
})

display(chunk_stats.head(10))
print("Number of chunks:", len(chunks))
print("Average words per chunk:", round(chunk_stats["words"].mean(),
1))
print("Minimum words:", chunk_stats["words"].min())
print("Maximum words:", chunk_stats["words"].max())
```

	chunk_id	page	characters	words
0	0	1	349	53
1	1	2	198	30
2	2	3	486	69
3	3	4	853	137
4	4	4	885	144
5	5	4	798	126
6	6	4	827	127
7	7	4	799	118
8	8	4	632	92
9	9	5	894	135

```
Number of chunks: 52
Average words per chunk: 118.7
Minimum words: 9
Maximum words: 149
```

Embedding

In [ ]:
```
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"
EMBEDDING_DEVICE = "cuda" if torch.cuda.is_available() else "cpu"

embedding_model = HuggingFaceEmbeddings(
    model_name=EMBEDDING_MODEL,
    model_kwargs={"device": EMBEDDING_DEVICE},
    encode_kwargs={"normalize_embeddings": True}
)

sample_embedding = embedding_model.embed_query(
    "How does Apple organize teams for innovation?"
)

print("Embedding model:", EMBEDDING_MODEL)
print("Embedding device:", EMBEDDING_DEVICE)
print("Embedding dimensions:", len(sample_embedding))
```

```
modules.json: 100%  349/349 [00:00<00:00, 25.8kB/s]

config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 13.5kB/s]

README.md: 10.5k/? [00:00<00:00, 831kB/s]

sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 5.50kB/s]

config.json: 100%  612/612 [00:00<00:00, 64.2kB/s]

model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 75.7MB/s]

tokenizer_config.json: 100%  350/350 [00:00<00:00, 36.5kB/s]

vocab.txt: 232k/? [00:00<00:00, 11.2MB/s]

tokenizer.json: 466k/? [00:00<00:00, 28.4MB/s]

special_tokens_map.json: 100%  112/112 [00:00<00:00, 12.0kB/s]

config.json: 100%  190/190 [00:00<00:00, 24.9kB/s]

Embedding model: sentence-transformers/all-MiniLM-L6-v2
Embedding device: cuda
Embedding dimensions: 384
```

Vector Database

In [ ]:
```python
# Delete any existing collection before rebuilding it.

COLLECTION_NAME = "apple_hbr_innovation"

try:
    existing_store = Chroma(
        collection_name=COLLECTION_NAME,
        embedding_function=embedding_model
    )
    existing_store.delete_collection()
    print("Deleted existing Chroma collection.")
except Exception as exc:
    print("No existing collection needed deletion:", exc)

vector_store = Chroma.from_documents(
    documents=chunks,
    embedding=embedding_model,
    collection_name=COLLECTION_NAME,
    collection_metadata={"hnsw:space": "cosine"}
)

print("Chroma collection created.")
print("Expected chunks:", len(chunks))
print("Indexed chunks:", vector_store._collection.count())
```

```
Deleted existing Chroma collection.
Chroma collection created.
Expected chunks: 52
Indexed chunks: 52
```

Retriever

In [ ]:
```python
# Retrieve the four chunks most semantically relevant to each query.
TOP_K = 4

retriever = vector_store.as_retriever(
    search_type="similarity",
    search_kwargs={"k": TOP_K}
)

def inspect_retrieval(query: str, k: int = TOP_K) -> pd.DataFrame:
    """Return retrieved chunks with page and similarity-relevant tex
t."""
    retrieved = vector_store.similarity_search_with_relevance_scores(
        query,
        k=k
    )

    rows = []
    for rank, (doc, score) in enumerate(retrieved, start=1):
        rows.append({
            "rank": rank,
            "page": doc.metadata["page"],
            "chunk_id": doc.metadata["chunk_id"],
            "relevance_score": round(float(score), 4),
            "text_preview": doc.page_content[:240] + "..."
        })

    return pd.DataFrame(rows)

display(inspect_retrieval(question_1))
```

	rank	page	chunk_id	relevance_score	text_preview
0	1	3	2	0.7211	PHOTOGRAPHER MIKAEL JANSSON How Apple Is Organ...
1	2	1	0	0.6915	REPRINT R2006F PUBLISHED IN HBR NOVEMBER–DECEM...
2	3	2	1	0.6176	2 Harvard Business Review November–December 20...
3	4	11	51	0.6039	. MORTEN T. HANSEN is a member of Apple Univer...

Response Function

In [ ]:

```
QUERY_EXPANSIONS = {
    question_1: (
        "article title authors Joel M. Podolny Morten T. Hansen "
        "publisher Harvard Business Review publication"
    ),
    question_2: (
        "Apple leaders three leadership characteristics deep expertise
"
        "immersion in the details willingness to collaboratively debat
e"
    ),
    question_3: (
        "Apple leadership innovation example iPhone 7 Plus portrait mo
de "
        "dual-lens camera computational photography specialist teams "
        "collaborative debate attention to detail"
    ),
}

ANSWER_GUIDANCE = {
    question_1: (
        "State the two authors and the publisher directly and concisel
y."
    ),
    question_2: (
        "Use exactly three bullet points. Name each characteristic and
keep "
        "each explanation to no more than two lines."
    ),
    question_3: (
        "Explain concrete examples supported by the context. Treat the
"
        "dual-lens camera and portrait mode as parts of the same conne
cted "
        "iPhone 7 Plus innovation example rather than two separate exa
mples."
    ),
}


def retrieve_documents(question: str, k: int = 8):
    """Retrieve relevant, nonduplicate passages using an expanded quer
y."""
    retrieval_query = QUERY_EXPANSIONS.get(question, question)
    candidates = vector_store.similarity_search(retrieval_query, k=k)

    unique_documents = []
    seen_chunk_ids = set()

    for document in candidates:
        chunk_id = document.metadata["chunk_id"]
        if chunk_id not in seen_chunk_ids:
```

```
            unique_documents.append(document)
            seen_chunk_ids.add(chunk_id)

    return unique_documents


def format_context(documents_to_format) -> str:
    """Format retrieved passages with source labels supplied by metada
ta."""
    formatted_passages = []

    for source_number, document in enumerate(documents_to_format, star
t=1):
        page = document.metadata["page"]
        chunk_id = document.metadata["chunk_id"]
        formatted_passages.append(
            f"[Source {source_number} | PDF page {page} | chunk {chunk
_id}]\n"
            f"{document.page_content}"
        )

    return "\n\n".join(formatted_passages)


RAG_SYSTEM_PROMPT = """
You are a retrieval-augmented business-research assistant.

Answer only from the retrieved context. Do not use outside knowledge.
Do not invent facts, quotations, sources, citations, or page numbers.
Do not write page citations in the answer; the application will displa
y
verified source metadata separately.
If the context does not support the answer, state that clearly.
""".strip()


def rag_answer(
    question: str,
    max_tokens: int = 650,
    temperature: float = 0.0,
    k: int = 8,
    show_sources: bool = True
):
    """Retrieve report evidence and generate a source-grounded answe
r."""
    start_time = time.perf_counter()

    retrieved_docs = retrieve_documents(question, k=k)
    if not retrieved_docs:
        raise ValueError("No passages were retrieved for the questio
n.")

    context = format_context(retrieved_docs)
    guidance = ANSWER_GUIDANCE.get(
        question,
        "Answer clearly and concisely using only the retrieved contex
t."
```

```
    )

    prompt = f"""
System instruction:
{RAG_SYSTEM_PROMPT}

Retrieved context from {SOURCE_DOCUMENT}:
{context}

Question:
{question}

Answer requirements:
{guidance}
""".strip()

    answer = generate_text(
        prompt,
        max_tokens=max_tokens,
        temperature=temperature
    )

    elapsed = time.perf_counter() - start_time
    source_pages = sorted({
        document.metadata["page"] for document in retrieved_docs
    })

    if show_sources:
        print(answer)
        print(
            "\nVerified retrieved PDF pages: "
            + ", ".join(str(page) for page in source_pages)
        )
        print(f"\nRetrieved passages from {SOURCE_DOCUMENT}:")

        for rank, document in enumerate(retrieved_docs, start=1):
            excerpt = textwrap.shorten(
                document.page_content,
                width=260,
                placeholder="..."
            )
            print(
                f"{rank}. PDF page {document.metadata['page']}, "
                f"chunk {document.metadata['chunk_id']}: {excerpt}"
            )

        print(f"\nResponse time: {elapsed:.2f} seconds")

    return {
        "question": question,
        "answer": answer,
        "source_document": SOURCE_DOCUMENT,
        "source_pages": source_pages,
        "documents": retrieved_docs,
        "response_time_seconds": elapsed
    }
```

In [ ]:

```python
# Verify retrieval before generating final RAG answers.
for question in [question_1, question_2, question_3]:
    print("\n" + "=" * 100)
    print("QUESTION:")
    print(question)

    diagnostic_docs = retrieve_documents(question, k=10)

    for rank, document in enumerate(diagnostic_docs, start=1):
        print(
            f"\nResult {rank} — PDF page {document.metadata['page']},
"
            f"chunk {document.metadata['chunk_id']}"
        )
        print(document.page_content[:500])
```

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QUESTION:
Who are the authors of the article 'How Apple Is Organized for Innovation,' and who published it?

Result 1 — PDF page 11, chunk 51
. MORTEN T. HANSEN is a member of Apple University’s faculty and a professor at the University of California, Berkeley. He was formerly on the faculties of Harvard Business School and INSEAD. FOR ARTICLE REPRINTS CALL 800-988-0886 OR 617-783-7500, OR VISIT HBR.ORG Harvard Business Review November–December 2020 11 This article is made available to you with compliments of Apple Inc for your personal use. Further posting, copying or distribution is not permitted.

Result 2 — PDF page 11, chunk 50
. An intermediate step may be to cultivate the experts-leading-experts model even within a business unit structure. For example, when filling the next senior management role, pick someone with deep expertise in that area as opposed to someone who might make the best general manager. But a full-fledged transformation requires that leaders also transition to a functional organization. Apple’s track record proves that the rewards may justify the risks. Its approach can produce extraordinary result

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Result 4 — PDF page 3, chunk 2
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Result 5 — PDF page 6, chunk 18
. CEO Apple’s Functional Organization In 1997, when Steve Jobs returned to Apple, it had a conventional structure for its size and scope. It was divided into business units, each with its own P&L responsibilities. After retaking the helm, Jobs put the entire company under one P&L and combined the disparate departments of the business units into one functional organization that aligns expertise with decision rights—a structure Apple retains to this day. CEO 1998 6 Harvard Business Review November

Result 6 — PDF page 11, chunk 45
. (See the exhibit “Roger Rosner’s Discretionary Leadership.”) Now he guides and gives feedback to other team members so that they can develop software applica- tions according to Apple’s norms. Being a teacher doesn’t mean that Rosner gives instruction at a whiteboard; rather, he offers strong, often passionate critiques of his team’s work. (Clearly, general managers without his core expertise would find it difficult to teach what they don’t know.) The second challenge for Rosner involv

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Result 7 — PDF page 10, chunk 42
. Finally, as Apple’s product portfolio and number of projects have ex panded, even more coordination with other functions is required, incre asing the complexity of Roger Rosner’s Discretionary Leadership Appl e’s VP of applications, Roger Rosner, oversees a portfolio comprising four distinct categories that require varying amounts of his time and attention to detail. In 2019 it looked like this: Low expertise High e xpertise Highly involved in the details 30% 40% of time 15% 15% Not hi ghly involv

Result 8 — PDF page 1, chunk 0
REPRINT R2006F PUBLISHED IN HBR NOVEMBER–DECEMBER 2020 ARTICLE ORGANIZ ATIONAL CULTURE How Apple Is Organized for Innovation It’s about exper ts leading experts. by Joel M. Podolny and Morten T. Hansen This artic le is made available to you with compliments of Apple Inc for your per sonal use. Further posting, copying or distribution is not permitted.

Result 9 — PDF page 4, chunk 6
. Giving business unit leaders full control over key functions allows them to do what is best to meet the needs of their individual units’ c ustomers and maximize their results, and it enables the executives ove rseeing them to assess their performance. As the Harvard Business Scho ol historian Alfred Chandler documented, U.S. companies such as DuPont and General Motors moved from a functional to a multidivisional struct ure in the early 20th century. By the latter half of the century the v ast major

Result 10 — PDF page 10, chunk 41
. Rosner, the VP of applications, provides a good example. Like many o ther Apple managers, he has had to contend with three challenges arisi ng from Apple’s tremendous growth. First, the size of his function has exploded over the past decade in terms of both head count (from 150 to about 1,000) and the number of projects under way at any given time. C learly, he cannot dive into all the details of all those projects. Sec ond, the scope of his portfolio has widened: Over the past 10 years he has a

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QUESTION:
List the three leadership characteristics discussed in
'How Apple Is Organized for Innovation.' Use bullet points and explain each characteristic in no more than two lines.

Result 1 — PDF page 10, chunk 40
. When Apple was smaller, it may have been reasonable to expect leader s to be experts on and immersed in the details of pretty much everythi ng going on in their organizations. However, they now need to exercise greater discretion regarding where and how they spend their time and e fforts. They must decide which activities demand their full atten- tio n to detail because those activities create the most value for Apple. Some of those will fall within their existing core expertise (what the y still n

Result 2 — PDF page 7, chunk 25

. Together, all can strive to do the best work of their lives in their chosen area. Willingness to collaboratively debate. Apple has hundreds of specialist teams across the company, dozens of which may be needed for even one key component of a new product offering. For example, the dual-lens camera with portrait mode required the collaboration of no f ewer than Apple leaders are expected to possess deep expertise, be imm ersed in the details of their functions, and engage in collaborative d ebate.

Result 3 — PDF page 10, chunk 39
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Result 5 — PDF page 9, chunk 34
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Result 6 — PDF page 7, chunk 22
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Result 7 — PDF page 8, chunk 27
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view.

Result 8 — PDF page 8, chunk 26
40 specialist teams: silicon design, camera software, reliabil- ity en gineering, motion sensor hardware, video engineering, core motion, and camera sensor design, to name just a few. How on earth does Apple deve lop and ship products that require such coordination? The answer is co llaborative debate. Because no function is responsible for a product o r a service on its own, cross-functional collaboration is crucial. Whe n debates reach an impasse, as some inevitably do, higher-level manage rs weigh

Result 9 — PDF page 4, chunk 8
. Thus the key decision makers—the unit leaders—lack a deep understand ing of all the domains that answer to them. THE APPLE MODEL The compan y is organized around functions, and expertise aligns with decision ri ghts. Leaders are cross- functionally collaborative and deeply knowled geable about details. IDEA IN BRIEF Apple is ORGANIZATIONAL CULTURE CO PYRIGHT © 2020 HARVARD BUSINESS SCHOOL PUBLISHING CORPORATION. ALL RIG HTS RESERVED. 4 Harvard Business Review November–December 2020 This ar ticle is m

Result 10 — PDF page 7, chunk 24
. The difference is subtle, and executing on it isn’t simply a matter of a more complicated mathematical formula. It demands that Apple’s op erations leaders commit to extremely precise manufacturing toler- ance s to produce millions of iPhones and other products with squircles. Th is deep immersion in detail isn’t just a concern that is pushed down t o lower-level people; it is central at the leadership level. Having le aders who are experts in their areas and can go deep into the details has profou

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QUESTION:
Explain specific examples from the article where Apple's approach to leadership led to successful innovations.

Result 1 — PDF page 9, chunk 34
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Result 2 — PDF page 8, chunk 28
. Doing so is not always easy, of course. A leader’s ability to be bot h partisan and open-minded is facilitated by two things: deep understa nd- ing of and devotion to the company’s values and common purpose, an d a commitment to separating how right from how hard a particular path is so that the difficulty of execut- ing a decision doesn’t prevent it s being selected. The development of the iPhone’s portrait mode illust rates a fanatical attention to detail at the leadership level, intense collabor

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Result 6 — PDF page 5, chunk 9
WHY A FUNCTIONAL ORGANIZATION? Apple’s main purpose is to create produ cts that enrich people’s daily lives. That involves not only developin g entirely new product categories such as the iPhone and the Apple Wat ch, but also continually innovating within those categories. Perhaps n o product feature better reflects Apple’s commitment to continuous inn ovation than the iPhone cam- era. When the iPhone was introduced, in 2 007, Steve Jobs devoted only six seconds to its camera in the annual k eynote eve

Result 7 — PDF page 9, chunk 35
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Result 8 — PDF page 6, chunk 14
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phone with a more costly and better camera, the team would most likely have less credibility the next time it proposed an expensive upgrade o r featur

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In [ ]:

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for question in [question_1, question_2, question_3]:
    print("\n" + "=" * 100)
    print("QUESTION:")
    print(question)

    retrieved_docs = retrieve_documents(question, k=10)

    for rank, document in enumerate(retrieved_docs, start=1):
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QUESTION:
Who are the authors of the article 'How Apple Is Organized for Innovation,' and who published it?

Result 1 — Page 11, Chunk 51
. MORTEN T. HANSEN is a member of Apple University’s faculty and a professor at the University of California, Berkeley. He was formerly on the faculties of Harvard Business School and INSEAD. FOR ARTICLE REPRINTS CALL 800-988-0886 OR 617-783-7500, OR VISIT HBR.ORG Harvard Business Review November–December 2020 11 This article is made available to you with compliments of Apple Inc for your personal use. Further posting, copying or distribution is not permitted.

Result 2 — Page 11, Chunk 50
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Result 4 — Page 3, Chunk 2
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Result 5 — Page 6, Chunk 18
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Result 9 — Page 4, Chunk 6
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List the three leadership characteristics discussed in
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Dean, Apple University Morten T. Hansen Faculty, Apple University AUTH
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BR.ORG Harvard Business Review November–December 2020 3 This article i
s made available to you with compliments of Apple Inc for your persona
l use. Further posting, copying or distribution is not permitted.
```

Question Answering using RAG

Question 1: Who are the authors of this article and who published this article?

In [ ]:
```
rag_result_1 = rag_answer(
    question_1,
    max_tokens=300,
    temperature=0.0
)
```

```
The authors of the article 'How Apple Is Organized for Innovation' are
Joel M. Podolny and Morten T. Hansen. It was published by Harvard Busi
ness Review.

Verified retrieved PDF pages: 1, 2, 3, 6, 10, 11

Retrieved passages from HBR_How_Apple_Is_Organized_For_Innovation.pdf:
1. PDF page 11, chunk 51: . MORTEN T. HANSEN is a member of Apple Univ
ersity's faculty and a professor at the University of California, Berk
eley. He was formerly on the faculties of Harvard Business School and
INSEAD. FOR ARTICLE REPRINTS CALL 800-988-0886 OR 617-783-7500, OR VIS
IT...
2. PDF page 11, chunk 50: . An intermediate step may be to cultivate t
he experts-leading-experts model even within a business unit structur
e. For example, when filling the next senior management role, pick som
eone with deep expertise in that area as opposed to someone who might
make...
3. PDF page 2, chunk 1: 2 Harvard Business Review November–December 20
20 This article is made available to you with compliments of Apple Inc
for your personal use. Further posting, copying or distribution is not
permitted.
4. PDF page 3, chunk 2: PHOTOGRAPHER MIKAEL JANSSON How Apple Is Organ
ized for Innovation It's about experts leading experts. ORGANIZATIONAL
CULTURE Joel M. Podolny Dean, Apple University Morten T. Hansen Facult
y, Apple University AUTHORS FOR ARTICLE REPRINTS CALL 800-988-0886 O
R...
5. PDF page 6, chunk 18: . CEO Apple's Functional Organization In 199
7, when Steve Jobs returned to Apple, it had a conventional structure
for its size and scope. It was divided into business units, each with
its own P&L responsibilities. After retaking the helm, Jobs put the...
6. PDF page 11, chunk 45: . (See the exhibit "Roger Rosner's Discretio
nary Leadership.") Now he guides and gives feedback to other team memb
ers so that they can develop software applica- tions according to Appl
e's norms. Being a teacher doesn't mean that Rosner gives instruction
at...
7. PDF page 10, chunk 42: . Finally, as Apple's product portfolio and
number of projects have expanded, even more coordination with other fu
nctions is required, increasing the complexity of Roger Rosner's Discr
etionary Leadership Apple's VP of applications, Roger Rosner, oversees
a...
8. PDF page 1, chunk 0: REPRINT R2006F PUBLISHED IN HBR NOVEMBER–DECEM
BER 2020 ARTICLE ORGANIZATIONAL CULTURE How Apple Is Organized for Inn
ovation It's about experts leading experts. by Joel M. Podolny and Mor
ten T. Hansen This article is made available to you with compliments o
f...

Response time: 0.88 seconds
```

RAG Observation

The retrieved title-page evidence identifies by **Joel M. Podolny** and **Morten T. Hansen** and published by **Harvard Business Review** in the November–December 2020 issue.

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

In [ ]:
```
rag_result_2 = rag_answer(
    question_2,
    max_tokens=500,
    temperature=0.0,
    k=10
)
```

- **Deep Expertise**: Leaders are expected to have specialized knowled ge in their functions, allowing them to engage meaningfully in all wor k and make informed decisions.

- **Immersion in the Details**: Leaders should understand the intricac ies of their organization at least three levels down, facilitating qui ck and effective cross-functional decision-making.

- **Willingness to Collaboratively Debate**: Leaders must engage in op en discussions with peers, advocating for their views while being open to changing their minds based on evidence and collective input.

Verified retrieved PDF pages: 4, 6, 7, 8, 9, 10

Retrieved passages from HBR_How_Apple_Is_Organized_For_Innovation.pdf:
1. PDF page 10, chunk 40: . When Apple was smaller, it may have been r easonable to expect leaders to be experts on and immersed in the detai ls of pretty much everything going on in their organizations. However, they now need to exercise greater discretion regarding where and ho w...
2. PDF page 7, chunk 25: . Together, all can strive to do the best wor k of their lives in their chosen area. Willingness to collaboratively debate. Apple has hundreds of specialist teams across the company, doz ens of which may be needed for even one key component of a new produc t...
3. PDF page 10, chunk 39: . Meanwhile, the number of VPs approximately doubled, from 50 to 96. The inevitable result is that senior leaders h ead larger and more diverse teams of experts, meaning more details to oversee and new areas of responsibility that fall outside their cor e...
4. PDF page 6, chunk 16: . Now let’s turn to the leadership model unde rlying Apple’s structure. THREE LEADERSHIP CHARACTERISTICS Ever since Steve Jobs implemented the functional organi- zation, Apple’s managers at every level, from senior vice president on down, have been expecte d...
5. PDF page 9, chunk 34: . After months of engineering effort, a key s takeholder, the video engineering team (responsible for the low-level software that controls sensor and camera operations) found a way, and the collaboration paid off. Portrait mode was central to Apple’s...
6. PDF page 7, chunk 22: . That would dilute their collective expertis e, reducing their power to solve problems and generate and refine inno vations. Immersion in the details. One principle that permeates Apple is “Leaders should know the details of their orga- nization three leve ls...
7. PDF page 8, chunk 27: . However, given Apple’s size and scope, even the executive team can resolve only a limited number of stalemates. Th e many horizontal dependencies mean that ineffective peer relationship s at the VP and director levels have the potential to undermine not...
8. PDF page 8, chunk 26: 40 specialist teams: silicon design, camera s oftware, reliabil- ity engineering, motion sensor hardware, video engi neering, core motion, and camera sensor design, to name just a few. Ho w on earth does Apple develop and ship products that require such...
9. PDF page 4, chunk 8: . Thus the key decision makers—the unit leader s—lack a deep understanding of all the domains that answer to them. TH E APPLE MODEL The company is organized around functions, and expertise aligns with decision rights. Leaders are cross- functionally...
10. PDF page 7, chunk 24: . The difference is subtle, and executing on

```
it isn’t simply a matter of a more complicated mathematical formula. I
t demands that Apple’s operations leaders commit to extremely precise
manufacturing toler– ances to produce millions of iPhones and other...

Response time: 2.39 seconds
```

RAG Observation

The response is grounded in the passage that explicitly names all three characteristics. The retrieved evidence supports:

- **Deep expertise:** Apple places experts in charge of experts so decision-makers understand the technologies and professional work within their functions.
- **Immersion in the details:** Leaders are expected to understand important work several levels below them, enabling faster and better-informed decisions.
- **Willingness to collaboratively debate:** Leaders advocate strongly for evidence-based positions while remaining willing to change their views when another approach better serves Apple's shared purpose.

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

In [ ]:
```
rag_result_3 = rag_answer(
    question_3,
    max_tokens=750,
    temperature=0.0,
    k=10
)
```

One specific example of Apple's approach to leadership leading to succ
essful innovation is the development of the dual-lens camera with port
rait mode for the iPhone 7 Plus. This innovation required extensive co
llaboration among various specialist teams, including video engineerin
g, camera software, and camera sensor design, among others. The collab
orative debate among these teams was crucial, as it involved individua
ls from different functions who engaged in discussions, pushed back on
ideas, and built upon one another's contributions to arrive at the bes
t solution.

The leadership at Apple demonstrated a fanatical attention to detail a
nd a commitment to the company's values and common purpose, which faci
litated this collaborative environment. For instance, Paul Hubel, a se
nior leader involved in the portrait mode project, took a significant
risk by advocating for a feature that would enhance the camera's capab
ilities, knowing that its success was critical for the iPhone's appea
l. This risk paid off, as the portrait mode became a defining feature
of the iPhone 7 Plus, significantly influencing users' purchasing deci
sions and enhancing the reputations of Hubel and his team.

Additionally, the leadership model at Apple emphasizes accountability
without control, meaning that while leaders are responsible for the su
ccess of their projects, they do not have direct control over all cont
ributing teams. This approach can lead to a "good mess," where teams w
ork towards a shared purpose, as seen in the portrait mode project. Th
e successful outcome of this innovation illustrates how Apple's leader
ship fosters an environment conducive to collaboration and innovation,
ultimately resulting in products that enrich users' lives.

Verified retrieved PDF pages: 3, 5, 6, 7, 8, 9, 10

Retrieved passages from HBR_How_Apple_Is_Organized_For_Innovation.pdf:
1. PDF page 9, chunk 34: . After months of engineering effort, a key s
takeholder, the video engineering team (responsible for the low-level
software that controls sensor and camera operations) found a way, and
the collaboration paid off. Portrait mode was central to Apple’s...
2. PDF page 8, chunk 28: . Doing so is not always easy, of course. A l
eader’s ability to be both partisan and open-minded is facilitated by
two things: deep understand- ing of and devotion to the company’s valu
es and common purpose, and a commitment to separating how right from h
ow...
3. PDF page 7, chunk 25: . Together, all can strive to do the best wor
k of their lives in their chosen area. Willingness to collaboratively
debate. Apple has hundreds of specialist teams across the company, doz
ens of which may be needed for even one key component of a new produc
t...
4. PDF page 6, chunk 13: . It does, but in ways that differ from those
employed by conventionally organized companies. Instead of using overa
ll cost and price targets as fixed parameters within which to make des
ign and engineer- ing choices, R&D leaders are expected to weigh th
e...
5. PDF page 8, chunk 26: 40 specialist teams: silicon design, camera s
oftware, reliabil- ity engineering, motion sensor hardware, video engi
neering, core motion, and camera sensor design, to name just a few. Ho
w on earth does Apple develop and ship products that require such...
6. PDF page 5, chunk 9: WHY A FUNCTIONAL ORGANIZATION? Apple’s main pu
rpose is to create products that enrich people’s daily lives. That inv

```
olves not only developing entirely new product categories such as the
iPhone and the Apple Watch, but also continually innovating within...
7. PDF page 9, chunk 35: . It also requires those leaders to inspire,
prod, or influence colleagues in other areas to contribute toward achi
eving their goals. While Townsend is accountable for how great the cam
era is, he needed dozens of other teams—each of which had a long list
of...
8. PDF page 6, chunk 14: . It was a big wager that the camera’s impact
on users would be sufficiently great to justify its significant cost.
One executive told us that Paul Hubel, a senior leader who played a ce
ntral role in the portrait mode effort, was “out over his skis,”...
9. PDF page 10, chunk 39: . Meanwhile, the number of VPs approximately
doubled, from 50 to 96. The inevitable result is that senior leaders h
ead larger and more diverse teams of experts, meaning more details to
oversee and new areas of responsibility that fall outside their cor
e...
10. PDF page 3, chunk 2: PHOTOGRAPHER MIKAEL JANSSON How Apple Is Orga
nized for Innovation It’s about experts leading experts. ORGANIZATIONA
L CULTURE Joel M. Podolny Dean, Apple University Morten T. Hansen Facu
lty, Apple University AUTHORS FOR ARTICLE REPRINTS CALL 800-988-0886 O
R...

Response time: 3.44 seconds
```

RAG Observation

The retrieved passages connect Apple’s leadership model to a concrete product outcome. The strongest example is the continuing development of the iPhone camera and the creation of portrait mode for the iPhone 7 Plus. The article explains that portrait mode required collaboration across at least 40 specialist teams, intense debate over quality standards, a delayed release to address failure cases, and additional engineering work to provide a live preview. The feature became central to the product's marketing and a major reason customers selected and enjoyed the phone.

Evaluation of the RAG Prototype

The following lightweight evaluation checks whether the retrieved passages contain expected evidence and compares the baseline, prompt-engineered, and RAG approaches.

In [ ]:

```
evaluation_cases = [
    {
        "question": question_1,
        "expected_terms": [
            "Joel M. Podolny",
            "Morten T. Hansen",
            "Harvard Business Review"
        ],
        "k": 4
    },
    {
        "question": question_2,
        "expected_terms": [
            "deep expertise",
            "immersion in the details",
            "collaboratively debate"
        ],
        "k": 4
    },
    {
        "question": question_3,
        "expected_terms": [
            "portrait mode",
            "dual-lens",
            "collaboration"
        ],
        "k": 6
    }
]

evaluation_rows = []

for case in evaluation_cases:
    retrieved_docs = retrieve_documents(
        case["question"],
        k=case["k"]
    )
    retrieved_text = " ".join(
        doc.page_content.lower() for doc in retrieved_docs
    )

    matched_terms = [
        term for term in case["expected_terms"]
        if term.lower() in retrieved_text
    ]

    evaluation_rows.append({
        "question": case["question"][:65] + "...",
        "expected_terms": len(case["expected_terms"]),
        "matched_terms": len(matched_terms),
        "retrieval_term_recall": round(
            len(matched_terms) / len(case["expected_terms"]),
            2
        ),
        "matched_evidence": ", ".join(matched_terms),
        "retrieved_pages": sorted({
            doc.metadata["page"] for doc in retrieved_docs
```

```
        })
    })

evaluation_df = pd.DataFrame(evaluation_rows)
display(evaluation_df)

print(
    "Average retrieval term recall:",
    round(evaluation_df["retrieval_term_recall"].mean(), 2)
)
```

	question	expected_terms	matched_terms	retrieval_term_recall	matched_evidence	retriev
0	Who are the authors of the article 'How Apple ...	3	3	1.0	Joel M. Podolny, Morten T. Hansen, Harvard Bus...	
1	List the three leadership characteristics disc...	3	3	1.0	deep expertise, immersion in the details, coll...	
2	Explain specific examples from the article whe...	3	3	1.0	portrait mode, dual-lens, collaboration	[5

```
Average retrieval term recall: 1.0
```

Evaluation Observations

- **Baseline LLM:** Fast and convenient, but it lacks access to the assigned report and may omit or invent article-specific details.
- **Prompt engineering:** Improves formatting and encourages uncertainty disclosure, but prompting alone cannot provide missing source evidence.
- **RAG:** Supplies article passages before generation, produces page-level traceability, and is therefore the most appropriate approach for report-based business research.
- **Human review remains necessary:** Term-recall checks confirm that expected evidence was retrieved, but they do not fully measure factual faithfulness or answer quality.

Actionable Insights and Business Recommendations

Key Findings

1. **Retrieval materially improves factual grounding.**
   The baseline model may know general facts about Apple, but it cannot reliably prove that its answers came from the assigned report. RAG retrieves the relevant article passages before generating an answer, allowing the response to include page-level evidence.
2. **Apple aligns decision rights with functional expertise.**
   Product and technology decisions are made by leaders with deep domain knowledge rather than by general managers primarily responsible for individual product profit-and-loss results.
3. **The leadership model combines expertise, detail, and collaboration.**
   Apple expects leaders to possess deep expertise, remain immersed in operational details, and participate in evidence-based cross-functional debate.
4. **Leadership practices are connected to concrete innovation outcomes.**
   The article's portrait-mode example demonstrates how technical expertise, extensive cross-functional collaboration, rigorous quality standards, and willingness to delay a release contributed to a successful iPhone feature.
5. **Traceable retrieval is valuable for business analysts.**
   Displaying the page and excerpt behind each answer reduces the time required to verify findings and lowers the risk of relying on unsupported model output.

Business Recommendations

- Deploy RAG for long reports where analysts need fast answers supported by auditable source passages.
- Preserve document metadata, including page numbers, during ingestion so every answer can be verified.
- Use low generation temperature for factual research tasks to reduce unnecessary variation.
- Require the model to state when the retrieved context is insufficient rather than encouraging it to guess.
- Evaluate the system with representative analyst questions and manually verify whether the retrieved passages contain the expected evidence.
- Add document access controls, logging, versioning, and human review before using generated answers for high-impact investment or strategic decisions.

Prototype Limitations

- The prototype contains one report and does not yet test retrieval across a large document collection.
- PDF extraction quality depends on the document layout and may require OCR for scanned reports.
- The prototype depends on a paid external API, so cost, privacy, rate limits, and network availability must be considered.
- Retrieval relevance does not by itself guarantee that every generated sentence is faithful to the evidence.
- The system should therefore support source inspection and human validation.

Conclusion

The prototype demonstrates that semantic retrieval can convert an information-dense report into a searchable knowledge resource. Compared with direct LLM questioning, RAG provides more reliable, transparent, and efficient report interpretation by grounding answers in relevant passages from the source document.

Full-Code Submission Checklist

Before exporting the notebook:

1. Set the Colab hardware accelerator to **GPU** and confirm that a T4 is available.
2. Upload `HBR_How_Apple_Is_Organized_For_Innovation.pdf` with the exact filename used in the loading cell.
3. Add `OPENAI_API_KEY` to Colab Secrets.
4. Select **Runtime** → **Restart session**, then run all cells from top to bottom.
5. Confirm that the vector database reports **52 expected chunks and 52 indexed chunks**.
6. Confirm that no error or warning output remains and that all three RAG answers display verified retrieved pages.
7. Export the completed notebook as **HTML** and submit only that one `.html` file.

Power Ahead